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Mixed poloidal-toroidal fields in massive white dwarfs beyond barotropy

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ABSTRACT

Context. A white dwarf supported above the Chandrasekhar mass by a magnetic field needs most of that field to be toroidal, because the toroidal branch is the one that reaches the required energy. It also needs a poloidal component, or it has no exterior dipole and nothing for magnetic braking or an observer to act on.
Aims. We ask whether a self-consistent equilibrium can carry both at once, and identify precisely which assumption blocks it.
Methods. TBD.
Results. TBD – so far, the barotropic ceiling has been measured in this code: a self-consistent barotropic twisted torus on a 1.35 $M_\odot$ white dwarf saturates at $E_{\rm tor}/E_{\rm mag} = 0.04$.
Conclusions. TBD.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

TBD.

2. What the equilibrium forces, and what barotropy adds

It is worth separating the constraints, because they are usually quoted together and only one of them is negotiable.

Consider an axisymmetric star in magnetostatic equilibrium, with no flows, and write the field through the flux function $u = \varpi A_\varphi$,

$$\mathbf{B} = \frac{1}{\varpi}\,\nabla u \times \hat{\mathbf{e}}_\varphi + B_\varphi \hat{\mathbf{e}}_\varphi. \tag{1}$$

Neither ∇P nor $\rho\nabla\Phi$ has an azimuthal component, so the azimuthal component of the Lorentz force must vanish by itself. That component is proportional to $(\mathbf{B}_{\rm p}\cdot\nabla)(\varpi B_\varphi)$, so

$$\varpi B_\varphi = \beta(u), \tag{2}$$

that is, ϖB_φ is constant along each poloidal field line. Outside the star the field is in vacuum and B_φ vanishes, so β must vanish on every line that reaches the surface. Writing u_s for the largest value of u on the stellar surface, the toroidal field is confined to the closed-line region $u > u_s$. This is the twisted torus, and it is what our companion paper measures directly as a separatrix.

The point to make is that *none of this used barotropy*. Equation (2) and the confinement follow from axisymmetry and from the exterior being field-free. Generalising the equation of state does not release them.

What barotropy does add is a restriction on the poloidal source. If $\rho = \rho(P)$ then $\rho^{-1}\nabla P = \nabla h$ and the whole left-hand side of the Euler equation is a gradient, so $\rho^{-1}\mathbf{f}_L$ must be one too. Since the Lorentz force of Eq. (1) is everywhere parallel to ∇u, this forces $\rho^{-1}\mathbf{f}_L = M'(u)\nabla u$ for some function M, and the equilibrium collapses to the Grad–Shafranov equation (Grad & Rubin 1958; Shafranov 1966)

$$\Delta^* u = -4\pi\varpi^2\rho\, M'(u) - \beta(u)\beta'(u), \tag{3}$$

with the Bernoulli integral $h + \Phi - M(u) = \text{const}$. The source of the poloidal field is pinned to the form $\rho M'(u)$: it must be proportional to the density, with a coefficient constant on flux surfaces.

That single restriction is the subject of this paper. Relaxing it is what $\rho = \rho(P, s)$ buys, because then $\rho^{-1}\nabla P$ is no longer a gradient, the baroclinic term $\nabla\rho \times \nabla P$ is available to balance what the gradient cannot, and the poloidal source becomes free (Mastrano et al. 2011; Armaza et al. 2015). It is also the route by which Ciolfi & Rezzolla (2013) reach large toroidal fractions in relativistic stars, where the barotropic family had been limited to a few percent (Lander & Jones 2009; Ciolfi et al. 2009; Duez & Mathis 2010).

For a white dwarf the physical case for it is straightforward: a real white dwarf is stratified, in temperature and in composition, so barotropy is the simplification and not the realism. The price is that the equilibrium no longer determines itself – a stratification has to be specified, or derived and then checked for physical plausibility.

3. The barotropic ceiling, measured

Before building a solver to beat a ceiling, the ceiling should be measured in the same code, on the same star. The value usually quoted for the barotropic twisted torus is a few percent of magnetic energy in the toroidal component; what follows is that number obtained here.

Table 1. Converged barotropic twisted-torus configurations. Rows above the highest converged $B_{\rm ref}$ for each ζ are omitted because the iteration diverges rather than returning a worse equilibrium.

ζ	$B_{\rm ref}$ (G)	$E_{\rm tor}/E_{\rm mag}$	$\max\lvert B_\varphi\rvert$ (G)
1.0	10^{11}	0.004	9.84×10^{10}
1.0	3×10^{11}	0.029	2.97×10^{11}
1.0	10^{12}	0.040	7.23×10^{11}
1.1	3×10^{11}	0.029	3.10×10^{11}
2.0	10^{11}	0.003	1.02×10^{11}

The background is a field-free configuration at $\rho_c = 10^9\,{\rm g\,cm^{-3}}$ and $\mu_e = 2$, giving $M = 1.346\,M_\odot$, $R_{\rm eq} = 2.46\times10^8$ cm and $|W| = 2.257\times10^{51}$ erg. On it we solve Eq. (3) with $M'(u) = k_0$ constant, $k_0 = 10^{-13}$, and

$$\beta(u) = \beta_0\left(\frac{u-u_s}{u_{\rm norm}}\right)^{\zeta} \quad \text{for } u > u_s, \qquad 0 \text{ otherwise}, \tag{4}$$

with $u_{\rm norm}$ fixed from the purely poloidal solution so that the functional form does not drift as u evolves. The equation is nonlinear through $\beta\beta'(u)$ and is iterated to a relative change below 10^{-8}. We sweep β_0, quoted through the field it asks for, $\beta_0 = B_{\rm ref}\varpi_{\rm ref}$ with $\varpi_{\rm ref} = R_{\rm eq}/2$, and repeat for $\zeta = 1$, 1.1 and 2.

At $\beta_0 = 0$ the poloidal solution has $\max|\mathbf{B}_{\rm p}| = 2.45\times10^{12}$ G, and its closed-line region occupies 37.7% of the stellar volume. Space is therefore not what limits the toroidal component.

The sweep is summarised in Table 1. The highest converged toroidal fraction is

$$\left.\frac{E_{\rm tor}}{E_{\rm mag}}\right|_{\max} = 0.040, \tag{5}$$

at $\zeta = 1$ and $B_{\rm ref} = 10^{12}$ G, reached after 428 iterations. Steeper β profiles do worse: $\zeta = 1.1$ converges up to 0.029 and $\zeta = 2$ only to 0.003. Above those values the iteration leaves the basin entirely rather than degrading gracefully.

Two things follow. The first is that the few-percent figure quoted from the literature is reproduced here, on a white dwarf rather than a neutron star, which is worth knowing before it is used as an argument.

The second is quantitative and much sharper. At the ceiling the toroidal field holds $E_{\rm tor}/|W| = 1\times10^{-5}$. A $2\,M_\odot$ configuration of the kind our companion paper certifies requires $E_{\rm tor}/|W| = 0.203$. The barotropic twisted torus is not marginally short of what mass support needs; it is short by four orders of magnitude. Whatever supports a $2\,M_\odot$ white dwarf, it is not a self-consistent barotropic mixed equilibrium.

3.1. What this measurement does not settle

Two caveats, both real.

The density is held fixed. The sweep solves Eq. (3) on the field-free background rather than iterating the stellar structure alongside it. The confinement of Sect. 2 is geometric and survives that, and at these field strengths the back-reaction on the structure is small, but the ceiling quoted here is a ceiling at fixed density and should be called that.

The failure mode is a failure of the iteration. Above the values in Table 1 an under-relaxed fixed-point scheme stops converging. That is consistent with the equilibrium ceasing to exist, and it is what the literature reports, but the two are not the same statement, and separating them needs a continuation method that follows the branch rather than a sweep that restarts at each point. Until that is done, the honest claim is that the branch is unreachable here above 4%, not that it is empty.

4. The non-barotropic construction

TBD. The plan, stated so the next step is falsifiable:

1. Drop the requirement that $\rho^{-1}\mathbf{f}_L$ be a gradient, keeping Eq. (2) and the confinement, which Sect. 2 shows are not negotiable.
2. Choose the poloidal source freely rather than as $\rho M'(u)$, and solve for the pressure by integrating $\nabla P = \mathbf{f}_L - \rho\nabla\Phi$ along the resulting field, which is possible exactly when $\nabla\times\mathbf{f}_L = \nabla\rho\times\nabla\Phi$.
3. Measure the stratification this demands – the spread of P over surfaces of constant ρ – and ask whether it is plausible for a white dwarf, in magnitude and in convective stability. This is the step where the method can fail honestly, and it is the one worth doing first.
4. Only then ask how high $E_{\rm tor}/E_{\rm mag}$ and $E_{\rm tor}/|W|$ go.

5. Conclusions

TBD.

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